## 3.2.2 Reaction rates

| | Learning outcomes | Additional guidance |
|---|---|---|
| | *Learners should be able to demonstrate and apply their knowledge and understanding of:* | |
| **Simple collision theory** | | |
| **(a)** | the effect of concentration, including the pressure of gases, on the rate of a reaction, in terms of frequency of collisions | |
| **(b)** | calculation of reaction rate from the gradients of graphs measuring how a physical quantity changes with time | *M3.1, M3.2, M3.5*<br><br>Suitable physical quantities to monitor could include concentration, gas volume, mass, etc. |
| **Catalysts** | | |
| **(c)** | explanation of the role of a catalyst:<br>**(i)** in increasing reaction rate without being used up by the overall reaction<br>(ii) in allowing a reaction to proceed via a different route with lower activation energy, as shown by enthalpy profile diagrams | Details of processes are **not** required. |
| **(d)** | **(i)** explanation of the terms *homogeneous* and *heterogeneous* catalysts | |
| | (ii) explanation that catalysts have great economic importance and benefits for increased sustainability by lowering temperatures and reducing energy demand from combustion of fossil fuels with resulting reduction in $CO_2$ emissions | HSW9,10 Benefits to the environment of improved sustainability weighed against toxicity of some catalysts. |
| **(e)** | the techniques and procedures used to investigate reaction rates including the measurement of mass, gas volumes and time | **PAG9**<br>HSW4 Many opportunities to carry out experimental and investigative work. |
| **The Boltzmann distribution** | | |
| **(f)** | qualitative explanation of the Boltzmann distribution and its relationship with activation energy **(see also 3.2.1 c)** | *M3.1* |
| **(g)** | explanation, using Boltzmann distributions, of the qualitative effect on the proportion of molecules exceeding the activation energy and hence the reaction rate, for:<br>**(i)** temperature changes<br>(ii) catalytic behaviour **(see also 3.2.2 c)**. | *M3.1*<br><br>HSW1,2,5 Use of Boltzmann distribution model to explain effect on reaction rates. |